Umut Can Çubukçu

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EDUCATION

Sep 2022 – Jan 2027 **Sabancı University, B.S. in Computer Science and Engineering**
Minor in Mathematics | *Expected graduation: January 2027*

- **GPA:** 3.72/4.00 | High Honor in all seven graded semesters | Merit Scholarship, raised from 50% to 75% for academic performance
- **National entrance exam:** ranked 5,271st in the quantitative track of YKS 2022, among more than 3 million candidates
- **Relevant coursework:** Machine Learning, LLMs: Foundations and Practice, Artificial Intelligence, Digital Image and Video Analysis, Algorithms, Data Structures, Linear Algebra, Probability, Discrete Mathematics, Statistical Modelling, Numerical Analysis, Mathematical Analysis

RESEARCH EXPERIENCE

Apr 2026 – Present **Cost-Aware Batch-Mode Active Learning for Multi-Label Classification**
Supervisor: Prof. O. Erhun Kundakçıoğlu | *sole student researcher*

- Carried out the project end to end, from formulation and proofs to implementation and experiments. Formulated batch selection of instance–label queries under binary partial feedback as an optimization model with a hard annotation budget, batch-dependent cost (a one-time inspection charge per example plus a per-question charge) and a single pairwise redundancy coefficient across the example and label axes.
- Proved the problem NP-hard and showed that fixed-cardinality and budgeted models from prior work are special cases.
- Designed a GRASP + ALNS metaheuristic that comes within 2% of the Gurobi optimum in under 15 seconds; under the same budget, the model buys 18–48% more information than prior formulations on four benchmark datasets.

Sep 2026 – Present **Optimizing Review Effort under Statistical Guarantees with Small Language Models**
Supervisor: Prof. O. Erhun Kundakçıoğlu | *student team*

- Building grading support for an Operations Research course: small, locally hosted language models suggest rubric-based grades, a solver verifies student LP formulations, conformal prediction estimates the risk of each suggestion, and an optimization model allocates limited TA review time across submissions.

Feb 2026 – Jan 2027 **Graduation Project: LLM-Based Agent for Subgoal Generation and Active Information Acquisition**
Supervisor: Zeki Doruk Erden | *team of four*

- Developing an LLM-driven embodied agent in the AI2-THOR simulator that decomposes household tasks into subgoals; its Epistemic Critic decides before acting whether the agent knows enough, and otherwise chooses an information-gathering action guided by a commonsense belief over object locations.
- Built an evaluation that hides task-relevant objects at three levels and grades every critic decision against simulator ground truth; main experiments are in progress.

Sep 2025 – Jan 2026 **PURE: Camera Calibration for 3D Reconstruction and Gaussian Splatting**
Supervisors: Prof. Hüseyin Özkan, Prof. Erchan Aptoula

- Built a multi-view calibration and reconstruction pipeline with Structure from Motion (COLMAP), estimating camera intrinsics and poses for object-level, indoor and outdoor scenes.
- Trained 3D Gaussian Splatting reconstructions from the calibrated views and explored drone-based image collection for active view planning in the AirSim simulator.

Feb 2025 – May 2025 **PURE: Impact of AI-Assisted Reading Tools on University Learners' Critical Reading Skills**
Supervisor: Ozan Tekin, School of Languages

- Proctored the reading sessions and analyzed student responses with LDA topic modeling and BERT-based semantic analysis to compare engagement with texts with and without AI support.

Jun 2024 – Aug 2024 **PURE: Medium Access Control Layer Design for THz-Band Drone Networks**
Supervisor: Prof. Özgür Gürbüz

- Set up the ns-3/TeraSim simulation environment and led the simulation study of a MAC protocol for THz-band drone networks.
- Integrated a beamwidth-adjustment algorithm (C++) into the ADAPT MAC protocol and evaluated throughput and delay across network sizes and sector counts.
- Co-authored the final report and presented the work at the PURE poster session.

MANUSCRIPTS

2026 **Cost-Aware Batch-Mode Active Learning for Multi-Label Classification under Binary Partial Feedback**
U. C. Çubukçu, O. E. Kundakçıoğlu. Manuscript in preparation.

WORK EXPERIENCE

Sep 2026 – Present **Software Engineering Intern, Search – Hepsiburada, Istanbul**
- Product search with Elasticsearch; building a search prototype on the WANDS e-commerce relevance dataset.

Jul 2025 – Sep 2025 **Java Development Intern – Yaşar Bilgi**
Izmir, Turkey | *github.com/umutcanc7/Work-Scheduling-System*
- Developed a work scheduling and survey management system with Spring Boot, PostgreSQL, Liquibase and Docker, replacing a manual, spreadsheet-based attendance process.
- Implemented Keycloak authentication with role-based access control, built the admin panel in React + DevExtreme and set up scheduled e-mail notifications.

TEACHING EXPERIENCE

Sep 2025 – Jan 2026 **Learning Assistant, CS310 Mobile Application Development**
Sabancı University | *Instructor: Elif Dürdane Koç*
- Held office hours on Flutter and Dart, helped with projects, graded quizzes and app projects, proctored exams.

Jan 2025 – May 2025 **Learning Assistant, CS300 Data Structures**
Sabancı University | *Instructor: Elif Dürdane Koç*
- Explained topics and solved problems in recitations; held office hours on course material and homework.

SELECTED PROJECTS

Feb 2026 – Jun 2026 **ScoutRAG: Constrained Hybrid Retrieval with Grounding Verification**
CS455 Large Language Models | *team of three* | *github.com/umutcanc7/scoutrag*
- Combined symbolic filtering for hard constraints, LLM-based query mapping (local Llama 3.1 via Ollama), semantic re-ranking with sentence embeddings and a post-generation check that verifies every cited number against the retrieved rows; evaluated against structured-only and vector-only baselines.

Jan 2026 **Information-Theoretic Turkish Wordle Solver**
github.com/umutcanc7/TurkishWordleSolver
- Selects the guess that maximizes expected information gain (Shannon entropy over feedback patterns), with Wordle-accurate duplicate-letter handling; solves Turkish Wordle in 3.87 guesses on average; 99.25% of 100,000 simulated games are solved within six guesses.

Sep 2025 – Dec 2025 **Unsupervised Mitosis Detection in DIC Microscopy**
CS419 Digital Image and Video Analysis | *Prof. Erchan Aptoula*
- Designed an unsupervised pipeline with Otsu thresholding and morphological reconstruction for candidate extraction, and an interpretable morphology score fused with entropy-based weights that improved recall over uniform weighting.

2024 – 2026 **Other Projects**
- **Smile classification on CelebA** (CS412): compared fine-tuning strategies for a pretrained VGG-16 in PyTorch; 92.33% validation and 91.47% test accuracy. *github.com/umutcanc7/CS412_HW3*
- **Song2Vec** (CS455): Word2Vec next-song recommendation on 30Music playlists, evaluated with HR@K and NDCG@K, with cold-start and long-tail analysis. *github.com/umutcanc7/song2vec*
- **Sözcükçü**: offline Turkish word game for iOS and Android in Flutter, with a server-free deterministic daily word and unit-tested game logic. *github.com/umutcanc7/sozcukcu*
- **Full-stack e-commerce web application** (CS308; React, Node.js, MongoDB, Docker): Scrum Master of the team over five sprints.

SKILLS

Programming: Python, C++, Java, JavaScript, Dart, SQL, C, Verilog
ML, optimization, simulation: PyTorch, scikit-learn, pandas, NumPy, Gensim, sentence-transformers, OpenCV, Gurobi, ns-3
Software: Spring Boot, React, Node.js, Flutter, Firebase, Docker, Git, Elasticsearch, PostgreSQL, MongoDB, WebGL
Languages: Turkish (native), English (fluent)

SERVICE AND OUTREACH

Sep 2023 – May 2024 **Student Ambassador, Sabancı University**
- Guided campus tours for prospective students and their families.

Dec 2020 – Mar 2021 **Volunteer Mathematics Tutor, ÇYDD**
- Taught mathematics to children with limited access to education during the pandemic.